

Route G2: Local Scattering with a Recoiling Quantum Detector

Bounded tree-level effective-theory result

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Abstract

A positive local interaction between the G1 field and a dynamical massive detector field permits a real change of Kaluza–Klein mode. The detector is localized by its quantum state, not pinned by an external potential. It carries the compensating compact momentum and ordinary recoil. We derive the vertex, threshold, cross section, wavepacket selection rule and energy balance. One engineering example has a nonzero conversion cross section. This is a collision of two systems, not coordinate-induced mass change or an elementary-particle identification. No external drive or mass fit is used.

1 One local action and its stability scope

Keep the G1 background $\mathbb{M}^{3,1} \times S^1$ with radius R , signature $(+ - - -)$, $\theta \sim \theta + 2\pi$ and $D_\theta \Phi = (\partial_\theta + i\alpha)\Phi$. Switch off its diagnostic external sources during scattering. Add a complex field X neutral under that connection:

$$S = \int d^4x R d\theta \left[|\partial_\mu \Phi|^2 - \frac{|D_\theta \Phi|^2}{R^2} - M_5^2 |\Phi|^2 + |\partial_\mu X|^2 - \frac{|\partial_\theta X|^2}{R^2} - M_D^2 |X|^2 - \kappa_5 |\Phi|^2 |X|^2 \right], \quad (1)$$
$$M_5 > 0, \quad M_D > 0, \quad \kappa_5 \geq 0. \quad (2)$$

The interaction is point-local in all five coordinates. A wavepacket of X will localize the apparatus around the circle. This is *state localization*, not a delta-function defect in the action, a prescribed classical background, or an infinitely heavy momentum sink. The X field has its own global $U(1)$ charge; net charge, not total particle plus antiparticle number, is exactly conserved.

Both kinetic terms have positive sign. The classical potential

$$V = M_5^2 |\Phi|^2 + M_D^2 |X|^2 + \kappa_5 |\Phi|^2 |X|^2 \geq 0 \quad (3)$$

has its minimum at $\Phi = X = 0$, and spatial-gradient energies are nonnegative. There is no tree-level condensate, tadpole or quadratic mixing to alter G1's free tower. This is a classical stability statement, not a theorem about a renormalized quantum effective potential. Dimensions are $[\Phi] = [X] = 3/2$ and $[\kappa_5] = -1$. The interacting theory is a five-dimensional effective field theory (EFT), not a proposed ultraviolet completion.

2 The vertex fixes the momentum transfer

Using the *same* normalized modes for both fields,

$$\Phi(x, \theta) = \sum_n \phi_n(x) \frac{e^{in\theta}}{\sqrt{2\pi R}}, \quad m_n^2 = M_5^2 + \frac{(n + \alpha)^2}{R^2}, \quad (4)$$

$$X(x, \theta) = \sum_l x_l(x) \frac{e^{il\theta}}{\sqrt{2\pi R}}, \quad \mu_l^2 = M_D^2 + \frac{l^2}{R^2}. \quad (5)$$

The four-mode overlap is exact:

$$\kappa_5 \int_0^{2\pi} R d\theta u_m^* u_n u_j^* u_l = \frac{\kappa_5}{2\pi R} \delta_{n+l, m+j} \equiv g \delta_{n+l, m+j}. \quad (6)$$

Thus one dimensionless coupling determines all tree vertices,

$$\mathcal{L}_{\text{int}}^{(4)} = -g \sum_{m, n, j, l} \delta_{n+l, m+j} \phi_m^\dagger \phi_n x_j^\dagger x_l, \quad g = \frac{\kappa_5}{2\pi R}. \quad (7)$$

No mode-specific conversion matrix is chosen. With covariant state normalization and $\langle f|S-1|i\rangle = i(2\pi)^4 \delta^4(P_f - P_i) \mathcal{M}$, the distinct-scalar process is

$$\phi_n + X_l \longrightarrow \phi_m + X_j, \quad j = l + n - m, \quad \boxed{\mathcal{M}_{\text{tree}} = -g}. \quad (8)$$

There is no identical-final-particle factor 1/2. The four-dimensional energy and momentum are conserved. Compact momentum conservation is equally explicit:

$$\frac{n+\alpha}{R} + \frac{l}{R} = \frac{m+\alpha}{R} + \frac{j}{R}, \quad \Delta p_{5,X} = \frac{n-m}{R} = -\Delta p_{5,\Phi}. \quad (9)$$

The canonical translation generator conserves $n+l$; adding the unchanged charged-particle holonomy shift gives the kinetic-momentum version above. The detector is the momentum receiver. Large gauge relabeling $(\alpha, n, m) \mapsto (\alpha+k, n-k, m-k)$ leaves every formula invariant.

3 Why heavy recoil opens the previously closed channel

G1's candidate $\phi_n \rightarrow \phi_m + \chi_{n-m}$, with a massless bulk χ , costs at least $|n-m|/R$ in the emitted mode. For $M_5 > 0$, $m_n - m_m < |n-m|/R$, so that decay is closed. The present process is *scattering*, with an incoming detector. For its $l=0$ component, write $d = n-m \neq 0$ and $\Delta = m_n - m_m > 0$. The detector's rest-energy increase is instead

$$\delta E_X = \sqrt{M_D^2 + d^2/R^2} - M_D = \frac{d^2/R^2}{\sqrt{M_D^2 + d^2/R^2} + M_D}. \quad (10)$$

It tends to $d^2/(2M_DR^2)$, not $|d|/R$, for a heavy detector. An exothermic channel at vanishing ordinary incoming kinetic energy exists precisely when

$$\Delta > \delta E_X, \quad (11)$$

$$\boxed{M_D > \frac{d^2/R^2 - \Delta^2}{2\Delta}}. \quad (12)$$

Squaring is legitimate because $M_D + \Delta > 0$. For the G1 massive tower, $0 < \Delta < |d|/R$, so the right side is positive. Equality is threshold only; nonzero incident kinetic energy may open a channel that fails this rest-limit condition. This is a finite-recoil mechanism, not an unrecorded supply of energy.

4 Cross section, thresholds and inverse process

For a component l with center-of-mass incoming momentum $p_i > 0$,

$$E_n = \sqrt{m_n^2 + p_i^2}, \quad E_l = \sqrt{\mu_l^2 + p_i^2}, \quad \sqrt{s_l} = E_n + E_l, \quad (13)$$

$$p_f = \frac{\sqrt{[s_l - (m_m + \mu_j)^2][s_l - (m_m - \mu_j)^2]}}{2\sqrt{s_l}}. \quad (14)$$

Always test $\sqrt{s_l} > m_m + \mu_j$ *before* using the square root; its product alone can also be positive below the pseudothreshold. Otherwise the physical open-channel cross section is zero.

For the open channel, integrate the invariant two-body phase space with flux $4p_i\sqrt{s_l}$:

$$\frac{d\sigma_l}{d\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2 s_l} \frac{p_f}{p_i}, \quad (15)$$

$$\boxed{\sigma_l = \frac{g^2}{16\pi s_l} \frac{p_f}{p_i}}, \quad v_{M,l} = p_i \left(\frac{1}{E_n} + \frac{1}{E_l} \right), \quad (16)$$

$$K_l \equiv \sigma_l v_{M,l} = \boxed{\frac{g^2 p_f}{16\pi E_n E_l \sqrt{s_l}}}. \quad (17)$$

This is the usual relativistic two-body normalization [1, 2]; the contact amplitude is isotropic at this order. Both σ_l and K_l have dimension -2 . Neither is a dimensionless probability. In an exothermic channel, σ_l grows as $1/p_i$ as $p_i \rightarrow 0^+$, while K_l has a finite limit. No flux exists at exactly zero relative velocity for defining σ by division. For the inverse process at the *same* s_l ,

$$p_i^2 \sigma_{n,l \rightarrow m,j} = p_f^2 \sigma_{m,j \rightarrow n,l} = \frac{g^2 p_i p_f}{16\pi s_l}. \quad (18)$$

Preparing the inverse channel at the same small numerical momentum 0.2 instead is a different experiment and can be below threshold; this does not violate detailed balance.

One predeclared engineering example

Take $R = 1$, $M_5 = 0.5$, $\alpha = 0.25$ from the G1 quarter-holonomy card and add $M_D = 5$, $g = 0.05$, $p_i = 0.2$. These are arbitrary common energy units, not measured particle data. Consider $n = 1, l = 0 \rightarrow m = 0, j = 1$:

Quantity	Value
(m_1, m_0)	(1.34629120, 0.55901699)
(μ_0, μ_1)	(5, 5.09901951)
Critical M_D from (12)	0.24146563
$\sqrt{s}$, p_f	6.36506416, 1.02175458
$\sigma_{1,0 \rightarrow 0,1}$	$6.27164081 \cdot 10^{-6}$
$K_{1,0 \rightarrow 0,1}$	$1.17224311 \cdot 10^{-6}$

The energy ledger is

$$m_1 - m_0 = 0.78727421, \quad \mu_1 - \mu_0 = 0.09901951, \quad (19)$$

$$T_{\text{out}} - T_{\text{in}} = 0.68825469, \quad T_{\text{in}} = 0.01877296, \quad (20)$$

$$T_{\text{out}} = 0.70702765. \quad (21)$$

Here T is the total *ordinary three-dimensional* kinetic energy of both particles; internal kinetic energy is already in their four-dimensional rest masses. At the event, $1.25 + 0 = 0.25 + 1$ in compact momentum units and $E_n + E_l = E'_m + E'_j$. The final ordinary kinetic energy receives the difference. No work reservoir has been omitted.

5 A localized detector is a state, not an external drive

Choose the finite, exactly normalized preparation

$$c_l = \mathcal{N} e^{-l^2/(4w^2)} e^{-il\theta_0}, \quad l = -4, \dots, 4, \quad w = 1.2, \quad \theta_0 = 0.7, \quad \sum_l |c_l|^2 = 1. \quad (22)$$

Finite support defines this particular state; it is not an uncontrolled replacement of an infinite Gaussian. At a common narrow ordinary momentum centered on $-\mathbf{p}_i$, its canonical angular wavefunction is

$$\psi_{\text{ang}}(\theta, t) = \frac{1}{\sqrt{2\pi R}} \sum_l c_l e^{il\theta - iE_l t}, \quad \int R d\theta |\psi_{\text{ang}}|^2 = 1. \quad (23)$$

It is localized initially and freely spreads; there is no potential holding it fixed. Its first angular moment is

$$\langle e^{i\theta} \rangle_t = \sum_l c_{l+1}^* c_l e^{i(E_{l+1} - E_l)t}. \quad (24)$$

This norm is a canonical one-particle localization diagnostic, not an assumed Lorentz-covariant local probability density. For comparison, the local normally ordered field-density profile in this narrow-momentum limit is proportional to

$$\langle : X^\dagger X : (\theta, t) \rangle \propto \left| \sum_l \frac{c_l}{\sqrt{2E_l}} \frac{e^{il\theta - iE_l t}}{\sqrt{2\pi R}} \right|^2. \quad (25)$$

It is not normalized to unit probability by itself. The heavy-field limit suppresses the distinction between the two profile shapes.

Exact first-order wavepacket statement

Let $d\Pi_a(p) = d^3p / [(2\pi)^3 2E_a(p)]$. For normalized four-dimensional envelopes $f(p)$, $h_l(k)$, form

$$|\text{in}\rangle = \sum_l c_l \int d\Pi_n(p) d\Pi_l(k) f(p) h_l(k) |\phi_n, p; X_l, k\rangle, \quad \int d\Pi_n |f|^2 = \int d\Pi_l |h_l|^2 = 1. \quad (26)$$

At first order, the outgoing amplitude for a specified (m, j) is

$$A_{mj}(p', k') = -ig c_{j-n+m} \int d\Pi_n(p) d\Pi_{j-n+m}(k) f(p) h_{j-n+m}(k) (2\pi)^4 \delta^4(p + k - p' - k'). \quad (27)$$

The coefficient is zero if its index is outside the preparation. Nonzero smooth envelopes around the open example shell give nonzero amplitude; ordinary localization is supplied by these envelopes, not a background force. A positive envelope choice near that shell avoids an accidental destructive cancellation. This proves existence of a genuine packet conversion at leading order, without treating a plane-wave cross section as a probability.

For conversion $m \neq n$, when detector recoil is unobserved and only outgoing m is selected, orthogonality of distinct $j = l + n - m$ gives

$$P_m^{\text{LO}} = \sum_j \int d\Pi_m(p') d\Pi_j(k') |A_{mj}|^2 = \sum_l |c_l|^2 P_{n,l \rightarrow m, l+n-m}^{\text{LO}}[f, h_l]. \quad (28)$$

It is not $|\sum_l c_l \mathcal{M}_l|^2$. Absolute encounter probabilities require the envelopes and encounter geometry; no such probability is claimed from the engineering card alone. Translating θ_0 changes phases but not (28): an incident single n mode is uniform around the circle. A dephased detector with the same $|c_l|^2$ gives the same inclusive conversion; that observable alone cannot certify localization. Coherent incoming Φ modes or coherent recoil post-selection would define a different experiment.

The precisely scoped numerical packet rate

The kinetic approximation uses component overlap integrals $I_l = \int dt d^3x n_\Phi(x, t) n_{X,l}(x, t)$ with normalized conditional spatial profiles. It gives

$$N_m^{\text{LO}} \simeq \sum_l |c_l|^2 I_l K_{n,l \rightarrow m, l+n-m}. \quad (29)$$

Only under a common-overlap approximation $I_l = I$ can one write

$$N_m^{\text{LO}} \simeq I K_{\text{packet}}^{(m)}, \quad K_{\text{packet}}^{(m)} = \sum_l |c_l|^2 K_{n,l \rightarrow m, l+n-m}. \quad (30)$$

Each component has its own s_l and velocity. The recorded K_{packet} is this rate coefficient, not a universal wavepacket cross section or a probability. The approximation need not hold during substantial spreading; (27) is the appropriate starting point then. All open final m are enumerated before defining a target rate fraction: a sufficient finite search interval is $|m + \alpha| \leq R\sqrt{s_l}$, followed by the exact threshold test. The elastic $m = n$ channel is included in the total collision rate. Its scattering cross section is not the total probability of staying in the original KK sector: that probability also contains the identity part of S and its interference, omitted from the connected amplitude (27). No G1 probe weight is reused as a scattering fraction.

For the stated preparation the common-overlap coefficients are $K_{\text{packet}}^{(0)} = 1.11933296 \cdot 10^{-6}$ and $K_{\text{packet}}^{\text{all}} = 1.85661445 \cdot 10^{-6}$. Their ratio 0.60288929 is a target fraction of *scattering rates under this prescription*, not a universal conversion probability. At $l = 0$ the open final branches are $m = -1, 0, 1$.

6 Verification, limits and next physical decision

The companion verifier checks circle overlap, independent energy-shell roots and phase space, thresholds, conservation laws, inverse processes and gauge relabeling. It also checks packet spreading, trace/dephasing identities and full open-channel enumeration. Reports preserve the source hash and every assertion. G2 passes 262/262 implementation checks on independent rerun; G1 passes its separate 331/331 regression. The finite packet has $|\langle e^{i\theta} \rangle| = 0.91640536$ at $t = 0$ and 0.48041872 at $t = 5$, with conserved norm and mean energy. This verifies a freely evolving localized preparation, not a localized vacuum.

The engineering amplitude is tree level. Weak g is not by itself a bound on the error of a five-dimensional EFT: a physical use requires a cutoff above all occupied and accessible momenta, matched higher operators, and loop/counterterm control. At nonzero coupling the exact interacting propagator has self-energy corrections and cuts. G1's free pole and probe formulae are the quadratic baseline, not all-orders identities of (1). No numerical EFT-error bar, radiative naturalness, background stabilization, Standard Model assignment or Route-F fit is certified.

Physical result. An incoming excitation can leave in another existing mass branch through a local collision. The detector takes the compensating compact momentum and ordinary recoil. Neither the shared bulk mass M_5 nor a coordinate convention changes. The simple G1 decay obstruction is bypassed by changing the *physical process*, not violating its inequality.

Bounded next options. The most direct continuation is a recoil-resolved experiment: test the correlation $j - l = n - m$ together with the predicted ordinary outgoing momentum. A finite-encounter calculation can specify four-dimensional packets and evaluate (27) if an absolute probability is needed. A driven background is a backup only, with work included in the Hamiltonian ledger. Stabilization and particle-content questions remain separate G3/G4 hypotheses; they do not reopen this conditional tree-level calculation.

References

- [1] Particle Data Group, *Kinematics*, Review of Particle Physics (2024), sections 49.5–49.5.1, equations (49.27)–(49.33), official review.
- [2] D. Tong, *Quantum Field Theory*, sections 3.4 and 3.6, author's lecture notes.